The Routing Deadlock: Challenges We Can No Longer Ignore

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AGENDA

Background

BGP Protocol Overview

- why it's so important
- how it has evolved over the past three decades

Core challenges #1

Security Vulnerabilities

- Route hijacking, leaks, source spoofing
- Lack of trust model

Core challenges #2

Management Complexity

- Human errors & misconfigurations
- Anomaly detection & recover

Core challenges #3

Scalability Bottlenecks

- Routing table explosion
- Protocol rigidity

Why BGP Matters: The Internet's "Foreign Policy"

What is BGP?

- Border Gateway Protocol (BGP) is the de facto inter-domain routing protocol
- Connects over 90,000 Autonomous Systems (ASes) globally
- Enables policy-driven routing decisions based on business relationships

Key Characteristics:

- Decentralized: No central authority controls routing
- Policy-based: Each AS defines its own routing strategy
- Scalable: Designed for the Internet's massive scale

How It Works (Simplified):

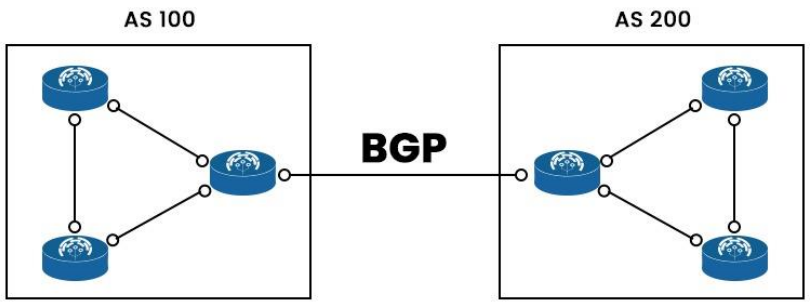
- ASes exchange reachability information via BGP UPDATE messages
- Path selection based on attributes: Local Preference, AS Path Length, MED, etc.
- Example: User in Asia accessing a cloud service in Europe

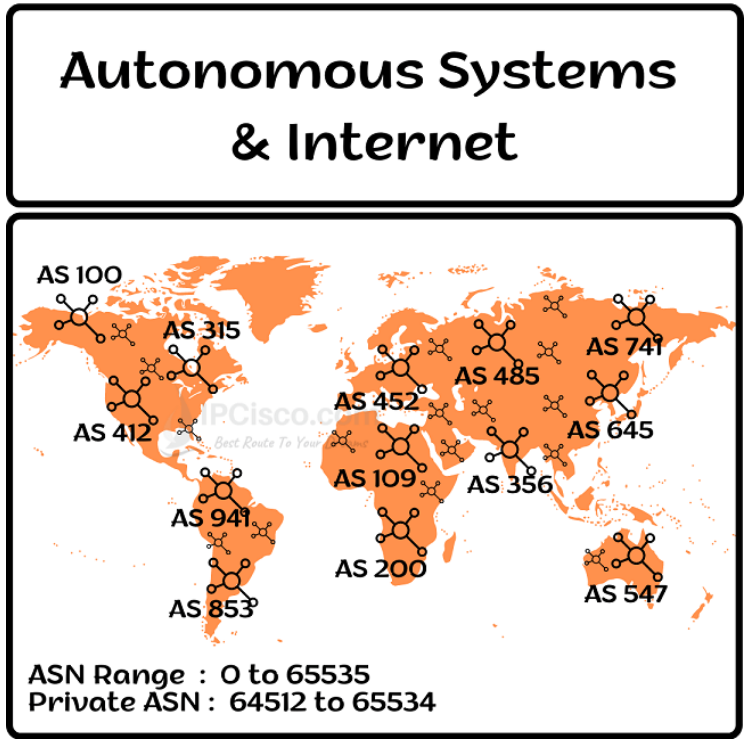
  → BGP selects optimal path through multiple transit providers

Why It Matters:

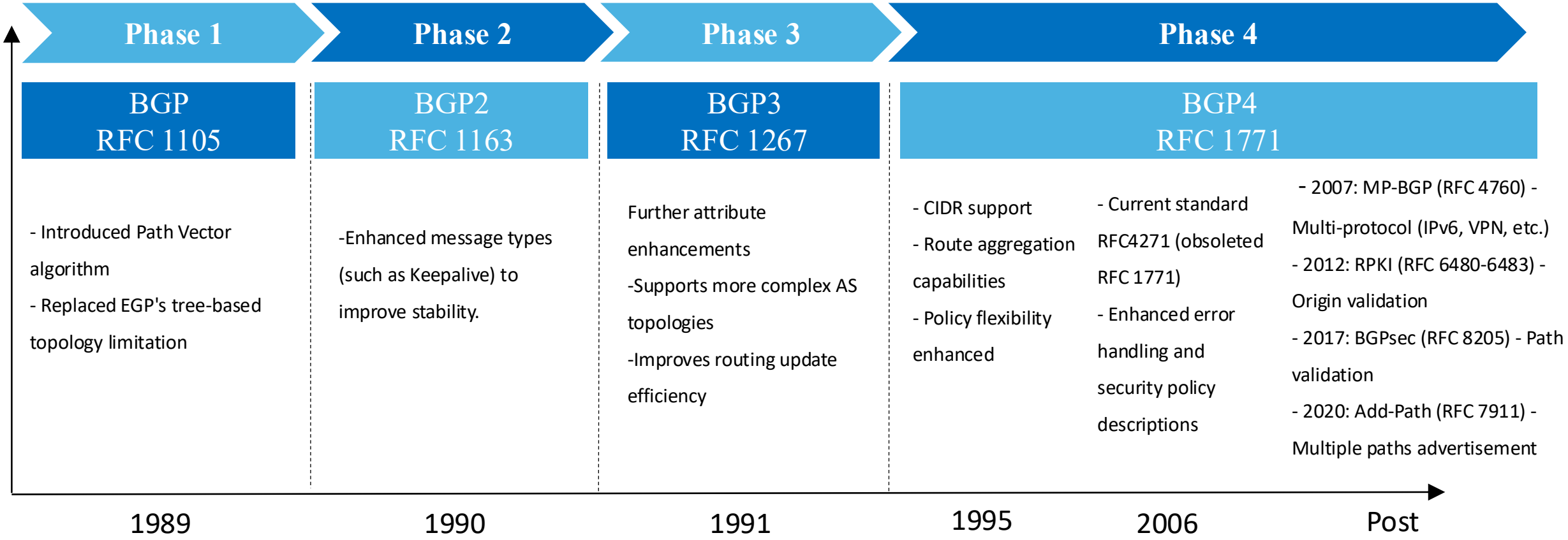
- BGP is the "glue" that holds the Internet together
- Every cross-AS communication depends on BGP
- Without BGP, the Internet would fragment into isolated networks

----**RFC References:** RFC 1771 (1995): BGP-4 standard , RFC 4271 (2006): Current specification





Evolution of BGP: 35+ Years of Continuous Adaptation



The Triple Challenge: Security & Management &Scalability

- **Security Vulnerabilities**
  - No built-in authentication
  - Route hijacking, leaks, spoofing
  - Trust model broken

- **Management Complexity**
  - Manual configuration prone to errors
  - Slow anomaly detection & recovery
  - AS coordination difficult

- **Scalability Bottlenecks**
  - Routing table > 1M entries
  - Hardware cost increases
  - Protocol rigidity limits innovation

- **The Root Cause:**

BGP's original design philosophy (1989) vs. Today's Internet demands (2025)
- 1989: Trusted environment, small network (~1,000 ASes)
- 2026: Adversarial environment, massive scale (~90,000+ ASes)

These challenges are not independent—they reinforce each other

Security Challenge : The Internet's "Unlocked Door"

BGP's Original Design Flaws (RFC 4272) which explicitly identifies three missing mechanisms:

- ☐ No message integrity/peer authentication
- ☐ No origin verification for NLRI announcements
- ☐ No path attribute authentication

Three Major Security Issues

1. Route Hijacking

Attackers forge route origins or alter path information in BGP UPDATE messages to hijack or redirect traffic maliciously.

Amazon Route Hijacking (2018)

Attackers forged route information, redirected DNS queries for cryptocurrency websites to fake servers, resulted in the theft of approximately 17 million USD worth of Ether.

2. Route Leak

Routes are propagated beyond their intended scope, violating routing policies. This can lead to eavesdropping, network congestion, or widespread outages.

Cloudflare Route Leak (2024)

BGP hijacking and route leak made Cloudflare 1.1.1.1 DNS server unavailable in 70+ countries.

3. Source Address Spoofing

Attackers forge the source IP address in packet headers to impersonate other entities.

Dyn DNS DDoS (2016)

The 2016 Dyn DNS attack, powered by the Mirai botnet, The attacker forges a large number of source IP requests to launch DDoS attacks.

The Missing Trust Model: Three Fundamental Tensions

TENSION #1: Distributed Trust vs. Centralized Certification

- **BGP's Design:** Peer-to-peer trust between neighbors "I trust my directly connected peers"
- **RPKI's Approach:** Hierarchical certification chain " IANA → RIRs → NIRs → LIRs → ASes"
- **The Conflict:**
  - Decentralized protocol meets centralized
  - trust anchors → Single point of failure
  - What if a RIR's database is compromised?

TENSION #2: Static Trust vs. Dynamic Networks

- **Current Model**: BGP sessions, once established, remain trusted indefinitely
- Reality:
  - Network conditions change
  - Peers may be compromised
  - Trust should be temporal and contextual
- **The Gap:**
  - No mechanism for dynamic trust adjustment
  - Can't revoke trust in real-time

TENSION #3: Privacy vs. Transparency

- **Current State**:
  - Routing policies are "black boxes"
  - ASes protect business sensitive information
- **The Problem:**
  - Opaque policies make collaboration hard
  - Can't verify if policies are compatible
  - Misconfigurations harder to detect
- **The Gap:**
  - Transparency helps security & management, but exposes business strategies to competitors

Management Challenge: The Human Dependency Problem

BGP Operations Today

- Manual configuration of route maps, filters, policies
- Human expertise required for troubleshooting
- Limited automation and intelligence
- AS coordination relies on out-of-band communication

CASE STUDY 1: Google Route Leak (August 25, 2017)

What happened: Google (AS15169) misconfigured BGP routes

Impact:

- NTT network outage in Japan
- Traffic incorrectly routed outside Japan
- Multiple critical services affected
- Duration: ~40 minutes

Root cause: Human configuration error

CASE STUDY 2: SafeHost Misconfiguration (June 6, 2019)

What happened: SafeHost (AS21217) misconfigured route updates

Scale:

- 70,000+ routes announced
- 368 million+ IP addresses affected
- Routes leaked

Impact:

- European traffic rerouted through Asia
- Severe performance degradation
- Duration: ~2 hours

The Root Cause:

BGP cannot validate route announcements → Any configuration error propagates globally

The Consequence: Human Errors → Global Impact

Management Challenge: Anomaly Detection & Recovery

The Current State

Detection

- Fragmented monitoring across ASes
- Each AS sees only local view
- No global real-time visibility

Recovery

- Multiple BGP UPDATE messages required
- Policy re-computation needed
- Inter-AS coordination delays
- Business disruptions during recovery

CASE STUDY: Facebook Global Outage (October 4, 2021)

What happened: During routine maintenance, Facebook accidentally:

- Withdrew BGP routes for its global backbone
- Disconnected data centers from Internet
- DNS servers became unreachable

Impact:

- WhatsApp, Instagram, Facebook down globally
- 6-hour complete outage
- Billions of users affected
- Economic impact: ~$100M estimated loss

Recovery timeline:

- Detection: immediate (internal)
- Diagnosis: ~2 hours
- Fix implementation: ~2 hours
- Full restoration: 6 hours

Key observation:

- Even a company with Facebook's resources
- took 6 hours to recover from BGP misconfiguration

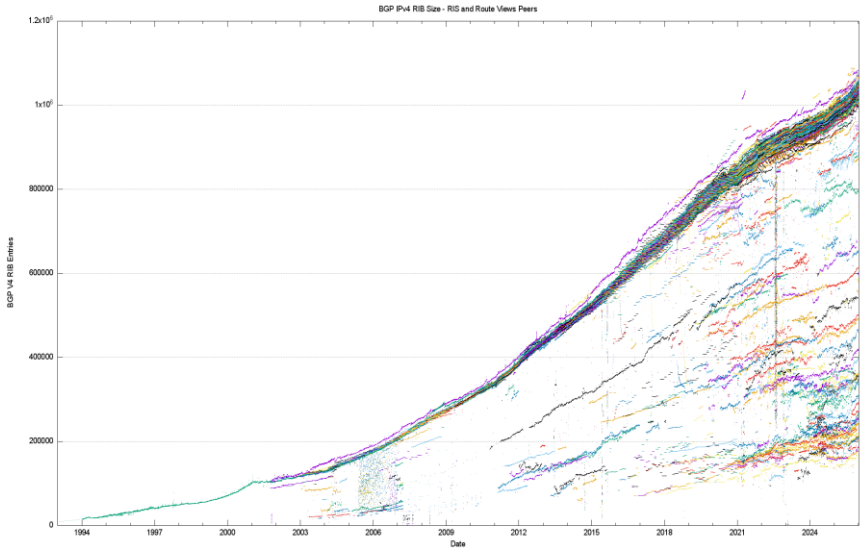
Why recovery is slow

- BGP's convergence algorithm is inherently slow
- No standardized inter-AS emergency coordination mechanism
- Manual intervention required for complex scenarios
- Cascading failures (BGP withdrawal → DNS failure →Service unavailable)

Scalability Challenge: When Growth Meets Protocol Limits

1. Routing Table Size & Performance

- Global BGP table has grown exponentially from tens of thousands to millions of entries. Larger tables consume significantly more memory and processing power on routers.
- Multi-homing and traffic engineering have significantly reduced CIDR aggregation effectiveness.Leads to slower convergence times and an increased risk of network instability.



IPv4 routing table size by RouteViews and RIS BGP peers

2. Protocol Rigidity & Adaptability

Slow and Complex Upgrades

The BGP protocol evolves slowly through the RFC process, making it difficult to quickly address new challenges.

Difficulty Supporting New Tech

BGP struggles to natively support modern networking concepts like Software-Defined Networking (SDN) and seamless IPv6 integration.

Feature Creep & Complexity

The protocol has accumulated many extensions over time, leading to increased complexity and potential for bugs.

The Next-Generation Routing System

- **We propose A SYSTEMIC REDESIGN of inter-domain routing**
  - Not incremental patches.
  - Not backward-compatible bandaids
- **But a CLEAN-SLATE architecture with:**
  - Built-in security (not add-ons)
  - AI-driven intelligence
  - Scalable beyond 1M routes
  - Open standards collaboration
- **The triple challenge demands it.**
- **The Internet's future depends on it.**
- **Let's build it together.**

Thanks!